Castles in the Air

Extended chapter commentary

CHAPTER 1
The noble art of proof

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CHAPTER 1 COMMENTARY

The noble art of proof

We have begun our story with a classic result from number theory: Euclid's theorem on the infinitude of the primes.

I should start by explaining why this is, because this book[a] isn't really about number theory. This book is really about a subject called *mathematical logic*, and about a bunch of questions that lie on the border between mathematics and philosophy. But in order to approach these questions — to set the scene — it's as well to start by looking at some actual examples of mathematics, so that we have a clear idea of what we're talking about. And number theory serves this purpose very well indeed.

Actually, part of me wishes this book *were* about number theory, because there's so much in number theory that would be fun to write about — but then I wouldn't be the person to write it. We'll be looking at a bit more number theory in Chapters 2 and 3, then embarking on mathematical logic proper from Chapter 4 onwards. But for the rest of this chapter, I'd like to expand on some of the ideas that have come up in our story so far, as a way of introducing the questions that will form the main topic of the book. In particular, we'll start by looking more closely at the crucial concept of a mathematical *proof*, and at the related idea that mathematical knowledge is in some way *certain*. As we shall see, these ideas lead us into a host of questions concerning the very nature of mathematics — questions that still perplex mathematicians and philosophers today. My hope is that by the end of the chapter, you'll have a pretty good idea of the sorts of things that this book is going to be about.

* * * * *

What is mathematics? Many people, I find, have a pretty definite idea of what being a biologist or an astronomer might involve, but will admit to being rather hazy about what it is that mathematicians spend their time doing. They sometimes vaguely suppose that it must mainly consist of doing enormously long and complex calculations — a bit like the ones they were made to do at

a We'll use the word 'book' here in a slightly generalized sense, covering the commentaries and other supplementary material on the website.

school, perhaps, only much bigger and scarier. Of course, that's not really what it's about: in fact, mathematicians are usually just as keen to avoid gruesome, tedious calculations as anyone else. So what *do* they do?

As one might expect, mathematics is too much of a vast, complex, varied, dynamic and evolving beast to be caged in by any simple, tidy description, but let me give at least a partial answer here, to be adjusted and extended later on. My first take on this will admittedly be rather biased to what is often called *pure* mathematics — broadly speaking, mathematics practised primarily for its own sake rather than with a view to applications. We will have more to say on the subtle and surprising interplay between mathematics and its applications in later chapters.

You'll get an impression of some important features of mathematics if you try picking up an advanced pure mathematics textbook or research paper and casually leafing through it at random. Your first reaction might be to feel dazed by the bizarre assortment of unfamiliar symbols and squiggles that confront you. But once your eyes have adjusted to this, you're likely to find that a lot of the text is organized into brief sections or paragraphs, each labelled with a heading: *Definition 4.2.1* … *Example 4.2.2* … *Proposition 4.2.3* … *Lemma 4.2.4* … *Theorem 4.2.5* … And within some of these sections, you will also see that the main portion of the text is headed by the all-important little word: PROOF.

Now, mathematics texts don't always adhere rigidly to this rather austere format, and I'm not saying that this is necessarily the best way to present mathematics. My point, rather, is that even at a superficial glance, this style reveals something important about what it is that mathematicians are typically trying to achieve. They are trying to construct an edifice of knowledge: an edifice built up brick by brick, with each stone resting firmly and securely on those below. The foundations of this edifice are definitions and other basic assumptions; the major blocks are statements known as propositions and lemmas; and the crowning pinnacles are the major results or theorems. So whether their goal is to attain a single towering new result or to develop the more basic infrastructure for a whole area of mathematics, we can think of mathematicians as builders — builders who want their building to be not be just a huge pile of stones, but something with an architecture and an intelligible shape.

What about the mortar that binds the bricks together? Here we come to one of the key defining characteristics of mathematics. For the edifice of mathematics is held together by nothing but pure, unalloyed logic. Each statement in mathematics leads inevitably to the next — not just with overwhelmingly high probability, not just by the general consensus of experts in the field, but by sheer logical necessity. And the process of building mathematical statements one upon another by means of such logical steps is what is called proof.

Proof in mathematics

What *is* a mathematical proof? Actually, a totally precise answer to this question isn't so easy to give, and that's one of the things this book will be about. But the basic idea is clear enough: a proof is just a logically watertight argument that shows beyond all doubt why some mathematical statement is true. Typically, it's an argument used to establish a statement that could never be checked simply by routine calculation — a statement such as 'There are infinitely many prime numbers'.

In our story, Euclid and Emily went through the proof of this statement in detail, in an informal but hopefully convincing way. This particular proof has been something of a favourite in popular writings on mathematics, but it's such a good example that I have no regrets about repeating it. I've included it partly because it's so interesting in its own right, but mostly to give you a feel for what a proof is like. I would encourage you to take the time to understand this proof properly, if you haven't already. I don't want you to take the result on trust from me, or from Euclid, or from anyone else — I want you to have the experience of *seeing for yourself* why the theorem is true, because once you've had this experience, you will have tasted for yourself the essence of what mathematics is really about.

The concept of proof is something that mathematicians have always been exceptionally proud of. It is their glory and their obsession. In pure mathematics, a statement is only considered to be definitively, reliably known once a fully rigorous proof has been given. Once this has been done, the statement is called a *theorem*. Whenever a mathematician succeeds in proving a new theorem, he or she adds another block to the growing edifice of mathematical knowledge. As the astrophysicist Arthur Eddington put it, 'Proof is the idol before whom the pure mathematician tortures himself.'[1]

And as mathematicians never seem to tire of telling everyone else, proof in mathematics is something qualitatively quite different from what is counted as 'evidence' in other disciplines. A statement in mathematics might come to seem overwhelmingly *likely* because it's been checked in a very large number of cases, but that's not at all the same thing as a proof. A classic example is given by the famous conjecture, made by Christian Goldbach in 1742, that every even number bigger than 2 can be got by adding two prime numbers together (for instance, 42 = 19+23). A massive computer search coordinated in Portugal[2] has now checked that this is indeed true for all even numbers up to 4,000,000,000,000,000,000. There are other reasons, too, for thinking that Goldbach's conjecture is overwhelmingly likely to be true. But so far, no one has managed to prove it. So a pure mathematician will insist that there's no *guarantee* that tomorrow a computer won't discover some large even number that *isn't* a sum of two primes. An accumulation of evidence, however strong it might appear, isn't considered a substitute for proof — on the contrary, it's

almost as though the stronger the accumulated evidence for some statement, the more desperately mathematicians long to find a proof of it.

You might find it hard to imagine that a statement like this *could* ever be false, given the amount of 'experimental' evidence for it. If so, see what you make of our next example, which again concerns prime numbers.

One idea that came up in our story is that the way the prime numbers are distributed amongst the other numbers appears chaotic and unpredictable. But despite this, there is actually a fairly simple one-line formula, proposed by Carl Friedrich Gauss in 1849,[3] which gives a pretty good estimate for the total *number* of primes up to any given number n. The estimate isn't exact, but it's never too far off: for instance, the formula predicts that there are around 78,628 primes below 1,000,000, whereas there are actually 78,498. Now, it would seem from looking at particular examples that the estimate is always a bit on the high side — the number given by the formula seems consistently a bit higher than the true number of primes up to n. Indeed, from the figures available to him, Gauss could see that this was the case for all values of n up to 3,000,000.

So it would seem eminently reasonable to conjecture that Gauss's formula errs on the high side for *all* values of n. Indeed, the cumulative evidence for this looks very strong indeed: it has now been definitively confirmed for all numbers up to 10 quintillion (= 10^{19}), and there's even good reason to suppose that it holds all the way up to 10^{300} (though this itself hasn't been rigorously proved yet). Surely, all of this adds up to pretty decisive confirmation that the conjecture is true?

Yet the bottom line is that this particular conjecture turns out to be false! In 1914, J.E. Littlewood showed that for certain fantastically large values of n, the formula actually underestimates the number of primes.[4] Admittedly this is a somewhat extreme example (which is why I chose it); but it does give a feel for how misleading numerical evidence can sometimes be, and for why it is that mathematicians will settle for nothing short of total, rigorous proof.

Later in this chapter, we'll start to explore some of the interesting philosophical issues raised by the concept of mathematical proof. But before we come to that, it will be helpful to look briefly at the history of this concept, and how the idea of proof has developed through the ages.

The Greek gift

The idea of mathematical proof is really due to the ancient Greeks. The Babylonians, the Chinese, the Indians and others had developed powerful numeral systems and sophisticated ways of calculating with them, and had also noticed some recurring patterns in both arithmetic and geometry. But it was the Greek sage Pythagoras and his school who were seemingly the first to

hit on the principle that facts about *all* numbers, or *all* triangles, or whatever, could be securely established as universally true by means of a sequence of logical steps. By means of this insight, the Greeks were able to build up a sizeable edifice of number theory and (more famously) geometry — an edifice that still stands to this day. In effect, what they had hit upon was nothing less than a completely new *way of knowing things* — a reliable means of acquiring general knowledge which could then be applied in an unlimited number of particular situations. Considered as a milestone in the history of knowledge, the importance of this breakthrough seems hard to overestimate.

The Greeks took their mathematics very seriously indeed. Pythagoras himself founded a kind of mystical sect which taught that everything in the universe was governed by — perhaps even ultimately constituted by — the perfect harmony of numbers and their exact ratios. Plato believed that the contemplation of mathematics cultivated in the soul a love of goodness, harmony and rectitude, and that consequently, only those who had undergone extensive training in arithmetic, plane geometry, solid geometry and astronomy were fit to serve in government. (One wonders how this suggestion would go down among today's politicians.) And Archimedes — the greatest mathematician of antiquity — reputedly paid for his life by refusing to comply with the demands of invading Roman soldiers until he had solved the mathematical problem he was working on at the time. Some of these attitudes might seem strange to us nowadays, but it's perhaps not too hard to imagine how the first practitioners of mathematical reasoning could have thought they had truly stumbled upon the ultimate secret of the universe.[5]

The great figurehead and systematizer of Greek mathematics was Euclid, who lived around 300 BC. Euclid's principal achievement was to gather together the main mathematical ideas of his time and organize them into a coherent, logical structure, presented in the thirteen books of his mighty treatise, the *Elements*. This was devoted mostly to geometry, though it also included important results on number theory, such as the theorem on the infinitude of the primes that we've looked at.[6]

By any standards, the *Elements* must rank as one of the great cornerstones of human civilization. Apart from anything else, it remained unchallenged as the supreme reference work on geometry for well over two thousand years — indeed, it was still used in schools in the late 19th and even early 20th century, making it far and away the most widely read and influential textbook of all time. But even more importantly, perhaps, it set the pattern for the whole development of mathematics. It is ultimately Euclid we have to thank for the modern 'Definition-Proposition-Proof' style of reasoning we mentioned earlier, and the *Elements* epitomizes the ideal of a mathematical edifice *par excellence*.

The key principle of Euclid's work was to start from a small list of basic assumptions or *axioms*, which were statements such as 'Any two points may

be joined by a straight line' or 'Things which equal the same thing also equal one another'.[7] These were statements whose truth was taken to be *so* obvious and indisputable that they required no justification and could be accepted without proof. Building on these, proposition by proposition, Euclid showed how from these simple axioms one could prove a wealth of geometrical results whose truth is often very far from obvious. (An example is shown in Figure 1.)

What about the axioms that we start from? Is there any possibility of 'proving' these too, or do they simply have to be taken on trust? We shall return to this point in later chapters — but for the time being, the broad idea is that unless we're willing to accept *something* as a starting point, we're not able to reach any interesting conclusions. So the axioms serve as the foundation on which the rest of the edifice rests. And one of the remarkable things about mathematics is that time and time again, one finds that just a short, tidy list of axioms is all that's needed to lay the foundation for an enormously rich and complex logical structure.

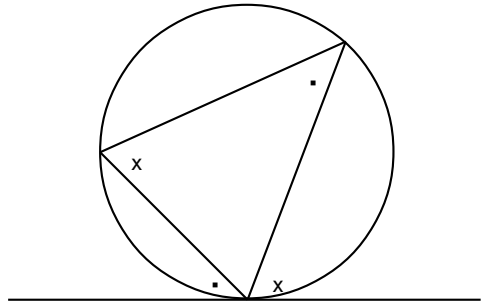


Figure 1: A non-obvious fact of Euclidean geometry (Proposition III.32 in the *Elements*): In any diagram of this form, the two angles marked with a dot are equal, as are the two angles marked with a cross. Euclid's proof of this depends directly on five previously proved propositions, which themselves build on earlier ones. In total, thirty earlier propositions are needed to support this proof.

This basic methodology — of building an edifice on a foundation of axioms by means of logical deductions — is nowadays known as the *axiomatic method*. Whilst it's not representative of the way mathematics has been done in all periods of history, since the mid-20th century[8] this approach has come to dominate the way pure mathematics is presented, and it's a way of thinking that today's mathematicians have deep in their bones. Of course, that's not to say that the process of mathematical *discovery* is anything at all like this steady, logical accumulation of truths: proofs and other ideas are often arrived at in a much more haphazard and experimental way involving creativity, trial-

and-error and sheer luck. But the finished result that most pure mathematicians aspire to is, I think, something like the kind of logical edifice we've been discussing. For this reason, Euclid deserves a place of high honour as one of the chief architects of mathematics as we know it today.

Has Euclid aged?

How well has Euclid's work stood the test of time? How does it stand up to scrutiny today? Pretty well on the whole, though there are a couple of important points to be made.

Firstly, it should be acknowledged that not all of Euclid's proofs are up to modern standards of mathematical rigour. It's not that the theorems themselves are wrong, but rather that his proofs occasionally smuggle in the use of additional assumptions which can't be deduced by logic alone from the basic axioms he declares up-front. Somewhat embarrassingly, one of these 'gaps' occurs in the proof of his very first proposition, which describes the construction of an equilateral triangle on a given base line. The construction consists of drawing two circles centred on the endpoints of the line, and taking the third corner of the triangle to be one of the points where these circles meet (see Figure 2). But how do we know there *is* a point where they meet? Looking at the diagram, this seems completely obvious — just as obvious, indeed, as the things Euclid does state as axioms. The issue, however, is that the existence of this intersection point doesn't follow *by logic alone* from Euclid's axioms — we need to fall back on what seems 'obvious from the diagram' in order to justify this step.

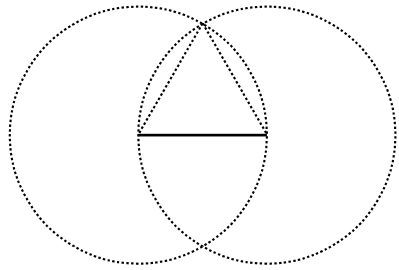


Figure 2: Euclid's construction of an equilateral triangle. The solid line segment is the given baseline; the dotted circles and lines show the constructions performed.

In fact, once we start putting the *Elements* under the microscope, we find that many of the proofs are positively riddled with what would today be seen as small gaps: unstated assumptions, or appeals to what 'the diagram' looks

like, or definitions that aren't precise enough. So today we would say that Euclid's axioms and definitions need to be supplemented with a few others in order to make all these extra assumptions explicit. (An axiomatic treatment of Euclidean geometry that achieved this was provided by David Hilbert in 1899, and an even better one by Alfred Tarski in 1926.[9]) So to a modern eye, the *Elements* appears rather like an ancient building that was constructed sufficiently well that it still stands today, and which still looks magnificent from a distance, but which on closer inspection turns out to suffer from numerous cracks, chinks and flaws in the masonry.

What would Euclid himself have said to all this? If these issues had been pointed out to him, would he have admitted that they were 'gaps' in his proofs? Or are we in danger here of trying to impose some ideal from a later era on an earlier one — of criticizing Euclid for not achieving something he was never aiming for in the first place? Is our identification of 'flaws' in the *Elements* really more a commentary on the modern mathematician's obsession with absolute logical hygiene than it is on the quality of Euclid's work — which, after all, served its purpose superbly for more than two millennia?

It is hard to be sure at this distance in time *exactly* what the ancient Greek geometers thought they were aiming for, and perhaps different schools of Greek thought held different views. However, given what we do know of their high ideals of pure logical reasoning and their stated rejection of appeals to mere plausibility or sense perception, I personally am inclined to think that Euclid *would* have quickly come to agree that there were gaps in his work, and that some further axioms ought to be added. At the same time, he would surely have been gratified to know that he was working along the right lines: that there was nothing *fundamentally* wrong with his basic approach, but that his own ideal of a logical edifice built on an axiomatic foundation would be further refined, clarified and perfected by the work of later mathematicians.

One geometry or many?

But there's also another, and perhaps more interesting, way in which our perspective today is different from Euclid's. Whereas Euclid would have referred to his subject simply as 'geometry' (perhaps pausing to make a distinction between plane and solid geometry), we would now say that what he was studying was just one of several possible 'geometries'. Thanks to the 19th century mathematicians Gauss, Bolyai and Lobachevsky, in addition to the so-called *Euclidean* geometry we now also have *hyperbolic* and *elliptic* geometry, which can be thought of loosely as the geometries applicable to various kinds of 'curved space'. In other words, whereas the ancient Greeks would have assumed that the basic laws of space were something fixed and 'absolute', we would now say that many different kinds of 'space' are

conceptually and mathematically possible, each characterized by a different set of basic axioms. Perhaps most surprisingly of all, we now know from Einstein's general theory of relativity that even the *physical* three-dimensional space we inhabit does not exactly satisfy the laws of Euclidean geometry, owing to the gravitational pull exerted by objects with mass — a discovery that would surely have astonished and fascinated the ancient Greeks.

I don't want to tell too much of this story here, since we'll be looking more in later chapters at these alternative geometries and the important issues they raise. For the time being, the important point is that rather than presenting Euclid's axioms or theorems as unconditionally 'true', we would nowadays say only that they are 'true within Euclidean geometry' — or better still, that the axioms in question *define* what Euclidean geometry is. Other geometries exist, and can be developed as logical edifices based upon suitable alternative sets of axioms. For example, in Euclidean geometry it can be proved that the angles of any triangle add up to 180° — or 'two right angles' as Euclid put it — whereas in elliptic geometry it can be shown that they always add up to more than 180°, and in hyperbolic geometry that they always add up to less. What's more, all of these different geometries can be accommodated comfortably alongside one another within the broader framework of modern mathematics, so that the existence of different geometries nowadays seems no more mysterious than the existence of different planets within our solar system. Euclid's world is simply one among many.

Maths lasts!

Despite these qualifications, however, one can't but be impressed overall by how well the mathematics of antiquity has lasted. Another thing mathematicians never seem to tire of telling the world is that whereas ancient Greek ideas on the composition of matter or the movements of the planets were discarded long ago, their mathematics has survived. As we have seen, their theorems on geometry are still accepted by mathematicians today, with the qualification that they apply to just one particular sort of geometry. Their results in number theory, on the other hand, are still accepted today without any such qualification. What's more, all this mathematical knowledge has been accepted universally and continuously from Euclid's time down to ours.

Having looked into some of the history, let's now start to reflect on all of this a bit more philosophically. How do we account for the remarkable degree of permanence and durability that mathematical knowledge seems to enjoy?

On one level, the answer to this seems obvious: mathematical knowledge is, or aspires to be, based on logically rigorous proof, and what is built by logic alone is in some way *certain* and therefore indestructible. Let's examine this very important idea more closely.

The lure of certainty

We here reach the mysterious heart of what appears to be special about mathematics. This, surely, is above all else what has drawn people to the subject through the ages. Mathematics, like no other discipline, seems to hold out a prospect of complete and utter certainty — the lure of something totally dependable and reliable. Questions in philosophy, politics, art or religion can be debated endlessly, and even scientific theories are in principle open to revision in the light of future discoveries; but mathematics, it would seem, is the one thing that it doesn't make any kind of rational sense to doubt. This is an enormously potent idea, and one that has gripped many minds down the centuries:[10]

> From all this we concluded: that the first two divisions of theoretical philosophy [i.e. theology and physics] should rather be called conjecture than knowledge … and that only mathematics can provide sure and unshakeable knowledge to its devotees, provided one approaches it rigorously. For its kind of proof proceeds by indisputable methods, namely arithmetic and geometry.
>
> — Claudius Ptolemy, *Almagest I.1* (c. 150)

> Doubt, then, what to hold for certain, sharply gnawed my heart … I wished to be as assured of the things I did not see, as I was that seven and three are ten, for I was not so far out of my mind as to think that even this could not be known; but I desired to have other things as clear as this.
>
> — Augustine of Hippo, *Confessions VI.4* (c. 397)

> Mathematics embraces arithmetic, plane geometry and solid geometry … They are matters of demonstration which it is impossible to deny once they have been understood and apprehended … Every student of mathematics admires its precision and the clarity of its demonstrations.[11]
>
> — Al-Ghazālī, *Deliverance from error* (c. 1100)

> It is confessed amongst all men who know what learning means, that besides the Mathematical arts there is no infallible knowledge, except it be borrowed from them.
>
> — Robert Recorde, *The whetstone of witte* (1557)

> We may safely conclude that physics, astronomy, medicine, and all the other disciplines which involve the study of composite things are indeed doubtful; but that arithmetic, geometry, and other disciplines of the same kind … contain something certain and indubitable.
>
> — René Descartes, *First Meditation* (1641)

> The charm [of mathematics] lies chiefly in the absolute certainty of its results; for that is what, beyond all mental treasures, the human intellect craves for. Let us be sure of something!
>
> — Charles Dodgson, a.k.a. Lewis Carroll, *A New Theory of Parallels* (1890)

> One reason why mathematics enjoys special esteem, above all other sciences, is that its laws are absolutely certain and indisputable, while those of other sciences are to some extent debatable and in constant danger of being overthrown by new discoveries.
>
> — Albert Einstein, *Geometry and Experience* (1921)

> The final result was a joint paper by Penrose and myself in 1970, which at last proved that there must have been a big bang singularity provided only that general relativity is correct and the universe contains as much matter as we observe. There was a lot of opposition to our work … However, one cannot really argue with a mathematical theorem. So in the end our work became generally accepted and nowadays nearly everyone assumes that the universe started with a big bang singularity.[12]
>
> — Stephen Hawking, *A Brief History of Time* (1988)

Necessary truth

There is also another powerful idea closely related to this, and which is perhaps another way of getting at essentially the same thing. This is the intuition that mathematical truths are in some way *necessary* truths. It's not just that 7+3 *happens* to be 10 — in some way or other, it *has* to be 10, and couldn't possibly have been anything else. We can imagine a world in which the sky is green, or one in which water flows uphill, or perhaps even one in which space has 943 dimensions. But can we coherently imagine a world in which 7+3 is 11? Or one in which there are only finitely many prime numbers? What would we even mean by that? Mathematics, it would appear, is in some way concerned with those things that *could not be otherwise*. As the philosopher Bertrand Russell put it, 'Mathematics takes us … into the region of absolute necessity, to which not only the actual world, but every possible world, must conform.'[13]

We have to be a bit careful here in the case of geometry. As we've already noted, the basic principles of Euclid's geometry have turned out not to be 'necessary' truths in the way that once they seemed to be: we *can* imagine worlds in which the angles of a triangle don't always add up to 180°, and indeed physicists now tell us that even our own universe is one of them. But what is still a necessary truth is that *within any geometry satisfying Euclidean axioms*, the angles of any triangle will add up to 180°. So this theorem, stated

in its modern 'conditional' form, would not be invalidated by any possible world.

Actually, all this talk about 'possible worlds' might be less idle than it sounds. Nowadays, some physicists are willing to take seriously the possibility that there might indeed be other 'universes', many of them with physical laws and geometry quite different from what we're used to. In connection with such ideas, the cosmologist Max Tegmark has commented:

> We shouldn't expect our intuition to work when we ask really deep questions about the ultimate nature of reality … So it's very important for us scientists to not diss ideas just because they feel weird. Fortunately, our math doesn't have any inhibitions and we can still calculate all these things, even if they seem completely counterintuitive, and it's only through the math that we're able to actually deal with all these ideas.[14]

So if these other universes exist, we can't trust our intuition to tell us what they might be like, but we can trust mathematics — because however strange and counterintuitive they might be in all other respects, they cannot but conform to the 'necessary' principles of mathematics. Indeed, the very thought that *mathematics itself* might not apply to these universes would seem in some way nonsensical. This is a striking example of the level of trust we habitually place in mathematics.

How certain is certain?

As we shall see, all of this raises some intriguing philosophical questions. But before we consider these, we should pause to examine the premise. Are we *right* to ascribe this unique level of certainty and necessity to mathematics? In particular, can we really say that mathematics is *absolutely* certain, as some of the above-cited luminaries suggest?

Whilst the broad idea that mathematics offers some quite exceptional level of certainty seems pretty hard to argue with, I personally would be wary of pushing this idea *too* far. Indeed, one can think of various ways in which the idea of mathematical certainty seems to require at least some qualification. Let me briefly touch on three of these here: one philosophical, one historical, and one sociological.

First, one might feel on very general philosophical grounds that there can never really be such a thing as 'absolute' certainty, just because there's nothing that couldn't be doubted by a sufficiently relentless and determined scepticism.[15] Let's try doubting our 'Infinitude of Primes' proof for just a moment. Perhaps there's a mistake in the proof somewhere which has miraculously escaped notice by you, me and all the millions of other people

who have checked it. Or perhaps each time you work through the proof you *do* find a mistake, but then some gremlin comes and modifies your memory so that you think it was all fine. Or perhaps… anyway, you get the idea. Of course, I'm not saying we ought to lose any sleep over these rather fatuous kinds of doubts — only that such thoughts make it a bit hard to see what one might mean by saying that mathematics was 'absolutely' certain. Furthermore, this sort of extreme doubting could equally well be applied to anything else we think we know — so these reflections don't in themselves seem to call into question the traditional idea that mathematics is the *most* certain kind of knowledge we have.

Secondly, even leaving aside such extreme scepticism, one might point out that the history of mathematics hasn't exactly followed the smooth, untroubled path that you might expect if all mathematical knowledge were absolutely 'certain'. As well as periods of relative stability and growth, there have also been times of crisis and upheaval during which mistakes have been uncovered, fundamental concepts redefined, former beliefs abandoned, and radically new perspectives adopted. The revolution caused by the discovery of non-Euclidean geometries is a well-known example of this; other classic examples include problems arising from the use of infinite series, or those stemming from the use of so-called 'infinitesimals' in differential calculus.

These historical episodes have important lessons to teach us, and they do indeed serve to correct the naive idea that mathematical activity is and always has been just a steady, gradual stockpiling of indestructible, certain knowledge. We will be looking in detail in later chapters at some of these crisis periods and what can be learned from them. But having said that, it's important not to overplay the significance of these episodes.[16] If there were 'crises' in 19th century mathematics that appeared to threaten the whole Euclidean enterprise, we have got over them. We have learned important lessons about the pitfalls of commonsense reasoning and the danger of relying on diagrams, and our confidence in the stability of our mathematical edifice is now stronger than ever before.

Besides this, it's possible to point to a significant portion of mathematics that has *never* in mathematical history been rocked by any such crisis or upheaval. No revolution in thought has ever cast any serious doubt on our basic conception of the sequence of Whole numbers (0, 1, 2, 3, …),[b] and there is a substantial corpus of results in number theory whose validity has never been in question, such as the beautiful theorem of Lagrange that every Whole number can be obtained as a sum of four perfect squares. (For instance, $6 = 2^2 + 1^2 + 1^2 + 0^2$, and $1111 = 31^2 + 11^2 + 5^2 + 2^2$.) So whilst history might lead us

b Throughout the book, I shall capitalize the word 'Whole' to indicate where it's being used as a technical term for a certain type of number. Likewise for 'Real' and 'Complex' numbers, when we get to them in Chapter 2.

to qualify our claims about mathematical certainty in some respects, it can equally well be said to support them in others.

Thirdly, there's an interesting line of thought which comes from reflecting on the sociological aspects of mathematical activity. The idea is this. Everyone agrees that human mathematicians are fallible, and that they might sometimes fail to spot a flaw in a mathematical proof. Now, in the case of, say, a two-page proof that has been carefully checked by, say, a hundred well-qualified mathematicians, the possibility that they might all have missed something might seem negligible. But what about a hundred-page proof that has been checked by just two mathematicians? Everyone who works in mathematical research is aware that such situations do arise: no one has the time to read and understand everything they'd like to, and once the proofs have been checked by a couple of expert referees and published, other mathematicians tend to assume the details are correct. But those two referees were only human, and were probably getting quite tired by page 82, so they could well have missed some subtle point. On the other hand, if a proof is unusually exciting and gets scrutinized and discussed by large numbers of people in university cafeterias around the world, our confidence in its correctness will be correspondingly boosted. In this sense, our level of faith in a proof will in practice depend partly on the *social process* to which it has been subject.[17]

This line of thinking has sparked some interesting debate. I think we should accept that the level of certainty we can claim for mathematical theorems is *in practice* limited by human fallibility. But one might also feel that we've wandered away from our original point here. Whilst we can certainly concede that the actual practice of mathematics is prone to error, this doesn't really seem to cast any doubt on the fundamental principles and methods of mathematics itself, if only we could isolate what these were. In other words, when mathematicians make mistakes, this isn't because there's anything doubtful about the methods themselves — it's just because they're not applying these methods in the way they mean to. Whether one *can* in fact precisely pin down these supposed principles of incontrovertibly correct mathematical reasoning will be an important theme of this book.

All three of these points, I think, provide valuable correctives to an overly simplistic idea of mathematical certainty. But they surely don't undermine the basic sense we all share that mathematical knowledge, or at least some of it, is *in some way* peculiarly reliable and certain — that there's something about a mathematical proof that compels acceptance by any rational mind. Exactly *which* parts of mathematics can claim this exalted status is a more complex and subtle question, and one that will occupy us for much of this book. However, if we restrict our attention to an area such as number theory, we really do find there a substantial corpus of precise, crystal-clear theorems that are not, in fact, disputed or doubted by anyone at all. Whilst there are some

sophisticated people around who might want to tell us that 'mathematical certainty' is an obsolete and discredited notion, it is in practice the case that even these people find a proof in number theory completely compelling when they see one. The doubts, or pseudo-doubts, seem to creep in only when we take our eye off what we're actually talking about. As mentioned earlier, Lagrange proved that every Whole number is a sum of four squares, and no one who understands the proof thinks there's any point at all in looking for one that isn't (even though, to put it mildly, you'd earn yourself instant fame and immortality if you found one). Once a theorem is proved, people stop arguing over it — because we do all feel, at bottom, that a mathematical theorem is something it would be *irrational to deny*. So it's pretty clear that there's something in the idea of mathematical certainty that we need to take seriously — even if the precise nature and extent of that 'certainty' are hard to pin down.

The view from outside

But now we come to a curious puzzle. *How* are our minds able to access such certain knowledge as we find in mathematics? What is the process by which we acquire mathematical knowledge? What is it about this process that makes the resulting knowledge is so utterly indubitable? Or why does it seem that way?

Philosophy has always been fascinated by questions like "How do we acquire knowledge?" and "Can we really know anything for sure?" So it's hardly surprising that mathematics — probably the closest thing we possess to a body of totally certain, reliable knowledge — should be of particular interest to philosophers.

So far, we've been discussing mathematics as it appears 'from the inside' — that is, from the perspective of someone actually doing it or thinking about it. But as a way of exploring these new questions, I'd invite you now to switch standpoint and imagine how the whole phenomenon of mathematics might look 'from the outside' — let's say, to a race of aliens visiting our planet.

We'll suppose, for the sake of argument, that the aliens possess a sophisticated technology and are skilled in basic arithmetic, but have no ability to understand general mathematical *proofs* in the way that we do. They are intrigued by our seemingly magical ability to know — just by going through some curious process within the comfort of our own skulls — that certain things hold true for *every single Whole number* (for instance, every Whole number is a sum of four squares). By means of simple calculations, the aliens are able to test many of these facts in particular cases (for instance, they can discover that $1111 = 31^2 + 11^2 + 5^2 + 2^2$), and they are appropriately impressed that this mysterious 'knowledge' of ours really does prove itself to

be outstandingly reliable in practice. So they're naturally led to ask: How do we do it? What is going on inside our human brains, biologically speaking, that gives rise to this knowledge? And what, if anything, about this process warrants the extreme level of *certainty* that we all ascribe to mathematics?

Of course, we don't (at present) have much detailed understanding of the specific brain processes that underlie mathematical thought. But it's not really the detail that I'm asking about here. The puzzle I'm wanting to highlight is a more philosophical one: What kind of process *could* it be that delivers this special kind of knowledge that is, in some way, beyond all rational doubt? From what source in the mind might mathematics spring?

Let's think about some of the ways in the aliens might try to account for this phenomenon. (Of course, the aliens are acting on our own behalf here, exploring some of the lines of explanation that we ourselves might propose.) First, they might try to explain it along broadly 'evolutionary' lines. For instance they might say that when we do mathematics, we're simply applying basic principles of reasoning which have always proved to be reliable in our experience of concrete situations, so we have good grounds for trusting them. Or, related to this, we might say that the innate reasoning principles with which evolution has endowed our minds have, by definition, passed the test of natural selection, so we can expect them to be very reliable.

But can an explanation along these lines come anywhere near explaining the levels of confidence that we actually place in mathematics? Ultimately, all such explanations boil down to saying: 'Well, it's worked well in an awful lot of cases, so it's probably right.' But we've already noted that the kind of confidence we gain from mathematical proof is — or at least, *seems* to be — something qualitatively quite different from that gained from a mere accumulation of examples. Besides, the trust that we actually place in mathematics extends far beyond those realms to which we or our biological ancestors have had any direct exposure. We have no qualms at all in applying mathematics to investigate numbers larger than the number of elementary particles in the observable universe, or weird alternative geometries that defy everyday intuition — or even the possible nature of other physical universes if they exist. What are we to make of this? After all, we're quite familiar with the idea that many of the faculties which which nature has endowed us (such as our ability to judge distances or probabilities) become very unreliable outside of the situations that they evolved to cope with. So why do we seem to make an exception for mathematics, affirming its rulings to be valid *beyond all rational doubt*, even in regions that far transcend our experience and stretch the bounds of our imagination?

One can put the question like this. Let's suppose that our aliens were fully acquainted with all of the evolutionary processes — biological, cultural and individual — that have given rise to our present mathematical faculties. Would this, in and of itself, lead the aliens to share *our* extreme level of

confidence in the universal and necessary validity of mathematics? Would they conclude from this that it was certain beyond doubt that there were infinitely many primes, or that every Whole number was a sum of four squares? Or would they say that these were merely working hypotheses, not to be trusted in realms in which they had not yet been tested? In the latter case, would the aliens conclude that our belief in mathematical certainty was somehow misplaced — or that our faith in rationality itself was founded on an illusion of some kind? Or is there something fundamentally different about the way our minds apprehend logical and mathematical truth? What is going on here?

We can also imagine a second scenario. Let's suppose now that some of the aliens *do* acquire the ability to grasp some basic principles of logical and mathematical reasoning. They might then carefully analyse our human mathematical proofs, and discover that they can all be accounted for as complex combinations of these basic principles. This might lead them to exclaim: 'Small wonder, then, that these human theorems always turn out to be true! They're bound to be, if they're based on *these* principles of reasoning, which are universally and necessarily valid by their very nature.'

Or, in a variant of this scenario, the aliens might undertake to monitor our brain activity in ultramicroscopic detail while we work through the proof of mathematical theorem, up to the point where we 'see', with unshakeable conviction, that it must be true. Let's suppose that the aliens actually succeed in understanding the brain processes at work, and that by means of their own mathematics — which by now has far outstripped ours — they're able to confirm that these brain processes will indeed lead to correct and reliable mathematical truths. In such a scenario, the aliens might even be able to reassure us that we're *right* to feel as certain as we do about mathematics.

I think it's clear, however, that in both these scenarios we have merely shifted the problem. What is going on in the *aliens*' brains (or equivalent) when they 'see' the validity of our basic reasoning principles, or apprehend the reliability of the underlying brain processes? How does reliable mathematical knowledge find its way into the physical universe at all? It might indeed be that both we and the aliens think we can just *see* that certain logical principles are valid, everywhere and for all time — but what physical process, whether in the aliens' brains or ours, could conceivably constitute a reliable verification of *that* stupendous fact?

These mystery would seem to be further compounded when we consider how much of mathematics involves the mysterious notion of *infinity* in some way. Even when we say 'Every Whole number is a sum of four squares', we seem to be invoking the notion of infinity in at least a mild form; but as we shall see later, there are many other concepts in mathematics that involve the concept of infinity in far more serious ways. But how are we able to *know* anything about infinity? What (presumably finite) process within our brains

could reliably be telling us anything at all about the infinite? Again, what conceivable brain process could ascertain that something holds true in infinitely many cases?

Could a digital computer ever perceive mathematical truth in the way that we can? Of course, computers can be programmed to generate proofs for large numbers of mathematical theorems, using logical rules supplied by us, and which *we* understand — but could a computer ever have that experience of 'seeing for itself' that the arguments it is using are logically valid, and hence that the resulting theorems 'must' be true? If so, could a computer be programmed to have a similar experience with arguments which we ourselves could see *weren't* valid?

At the end of the day, how *can* any purely physical process — whether within that marvellously structured grey porridge inside our skulls, or anywhere else — in and of itself produce mathematical knowledge that merits the cachet of 'indubitability' that we ascribe to it? *Why* do we put such supreme faith in mathematics?

At this point, we might perhaps waver (at least temporarily) in our conviction that mathematical knowledge *is* as certain as we suggested earlier. Perhaps our sense of mathematical certainty is an illusion of some sort, and we should simply learn to accept this? But can we really maintain such an attitude when we attend to what we are actually talking about? Can we *really* think it's on the cards that there might be a largest prime number, or a Whole number that's not a sum of four squares?

Perhaps mathematics is biologically so hard-wired into us that we *can't* doubt it, even if we want to? But we certainly can learn to doubt and question many of the other intuitions and instincts that evolution has bequeathed to us — so why should mathematics be different?

Perhaps mathematics is, in some strange way, just true *by definition* or by convention? Or perhaps it arises out of mind, language or human invention in some other way that makes the above questions misplaced? But can mathematics really be *entirely* a human invention or construction? Isn't there some sense in which it's not 'up to us' to decide whether or not the four-squares theorem hold? Surely, it's in some way *objectively true* that every Whole number is a sum of four squares — and somehow or other, we have a way of knowing that?

This sense of the startling mystery of mathematics is well expressed by the philosophers Alexander George and Daniel Velleman:

> Mathematics, in short, seems to be a discipline through which we acquire knowledge about infinitely many entities with which we can in no way causally interact, by means of finite inferences which make no use of empirical premises and which yield their conclusions with the force of necessity. We began by noting that it might seem surprising that

> mathematics should be the object of so much philosophical attention. But perhaps what should surprise is any display of intellectual equanimity before the phenomenon of mathematics.[18]

The basic puzzle, then, is this. 'From the inside', our mathematical knowledge seems — at least under favourable conditions — to be completely secure and not open to rational doubt. But 'from the outside', the reliability of any given process by which we might plausibly be acquiring this knowledge seems open to doubt in a thousand ways. How do we make sense of this? Must one of these viewpoints ultimately yield to the other? Or is there some perspective that does justice *both* to the certainty of mathematical knowledge *and* to the reality of how we acquire it? These are some of the philosophical questions that will lie behind our exploration.

What is mathematics *about*?

But these questions, perplexing though they may appear, are in one sense only the tip of the iceberg. For as soon as we start thinking about how we might acquire mathematical knowledge, we find ourselves coming up against some even more fundamental questions. What *is* mathematical knowledge anyway? What exactly is it that we know when we know a theorem in mathematics? What kind of meaning do mathematical statements have?

Let's return once more to Euclid's theorem: 'There are infinitely many prime numbers.' Actually, this is a very innocuous mathematical statement by comparison with many that we'll be considering later on; but even so, it will serve to illustrate many of the basic issues. We've satisfied ourselves by means of an unassailable mathematical proof that this statement is 'true'. But what does it actually *mean*? To put it very crudely: *Where* are there infinitely many prime numbers?

The very question sounds wrong, perhaps naive. Obviously, we're not claiming that an infinite list of prime numbers really 'exists' somewhere out there in ordinary physical space. But then, what *are* we saying? Is it perhaps *in our minds* that prime numbers exist? But surely no one thinks that anyone's mind can hold infinitely many numbers? Where then?

Perhaps we don't really mean that those infinitely many primes 'exist' anywhere at all? Maybe that's just a manner of speaking — a convenient shorthand for something else. But in that case, can we find some other way of explaining what that 'something else' might be?

For example, perhaps the reference to 'infinitely many' is a red herring. Perhaps all we really mean is: Given any *finite* list of prime numbers, there's some other prime number that's not on the list. (That, after all, was how Euclid phrased it.) But does this get round the problem? *Where* is there some prime number not on the list? In what sense might this new prime 'exist'?

More modestly still, perhaps we only mean that if we're presented with any finite list of primes, there's some *method for calculating* a prime not on the list? But again, where might there be a method for calculating this new prime?

Perhaps this *method* is something that could exist in our minds? This might seem more hopeful. But if so, then is Euclid's theorem ultimately just a statement about our own mental processes? Will this idea fly? How much of mathematics can be accounted for in this way?

If mathematics is ultimately *just* the study of things within our own minds, then why does it also seem to tell us so much about the outside world?

How much of mathematics *does* tell us something about the outside world anyway? For instance, does our statement 'There are infinitely many primes' say anything that might be experimentally tested or confirmed in some way? If it doesn't, does that raise a suspicion that it might not have any real meaning at all — let alone 'truth'? Could much of mathematics ultimately be nothing more than a curious linguistic game that mathematicians play?

Again, there's a strange kind of paradox here. On the one hand, if we focus on the *proof* of Euclid's theorem, we feel perfectly justified in saying we know, as certainly as we know anything, that there are infinitely many prime numbers: it all looks crystal-clear and there's nothing at all mysterious about it. But on the other hand, if we stop to ask ourselves what such a statement *means*, it's as though the ground suddenly drops away from beneath our feet and we find ourselves staring into an abyss. We possess — so it would seem — a piece of 'certain knowledge,' though we scarcely understand what it is we're certain *of*. As the cosmologist John Barrow put it, 'All our surest statements about the nature of the world are mathematical statements, yet we do not know what mathematics "is".'[19]

How do mathematicians themselves respond to these kinds of questions? What do *they* think mathematics is? As you might expect, many different views have been developed, and we'll be exploring quite a few of them in this book. But to get the ball rolling, I'd here like to introduce just one of them — perhaps the most familiar view of all, and certainly the one with the longest pedigree. This is what is known as the *platonist* view of mathematical truth, after the Greek philosopher Plato who flourished around 360 BC.

Plato's world

Take another look at Figure 1 on page 8. You will see that the diagram includes a circle. Thanks to the software I'm using, this hopefully looks like a pretty good circle, but we all know that it's not a *perfect* circle. The line has a definite thickness that a perfect circle wouldn't have, and whether you're looking at it on a screen or on paper, if viewed sufficiently close up it would doubtless appear grainy or bumpy in places. Likewise, the resolution limit of

my graphics software might mean that the corners of the triangle don't lie exactly on the intended circle, and this in turn might mean that the angles I've marked as equal aren't *precisely* equal in the diagram as printed. And so on.

But none of this seems to matter much, because we all understand that the theorem isn't really claiming anything about the particular diagram printed in your copy of this book, or about any other physical diagram anywhere else. Rather, the theorem is saying something that would be true in a 'perfect' or 'ideal' version of this diagram if one existed. And the curious thing here is that we all seem to have a very clear idea of what we mean by a 'perfect' version of the diagram, even though we've never seen one and we know that no such thing could ever be physically drawn.

Plato's view was that the entities that mathematics talks about don't exist physically, but they do 'exist' in a different way within a realm of 'ideas' to which our minds have access:

> You also know how [mathematicians] make use of visible figures and discourse about them, though what they really have in mind are the originals of which these figures are images: they are not reasoning, for instance, about this particular square and diagonal which they have drawn, but about *the* Square and *the* Diagonal; and so in all cases. The diagrams they draw and the models they make are actual things, which may have their shadows or images in water; but now they serve in their turn as images, while the student is seeking to behold those realities which only thought can apprehend.[20]

Nowadays, the term 'platonism' (or 'Platonism' if you prefer) is generally used somewhat loosely to cover a broad range of philosophical positions, not just Plato's particular theory of ideas.[21] But the core ingredient of platonism is simply the idea that there is some kind of objective truth or reality, independent of ourselves, to which mathematical statements refer. Just as there is (we commonly suppose) a physical world outside ourselves to which we refer when we speak of trees and elephants, so there is a realm of mathematical reality outside ourselves to which we refer when we speak of prime numbers or non-Euclidean geometries. This platonic realm of reality is typically held to be abstract, non-physical, timeless and immutable. What *kind* of existence or reality might such a realm be said to have? Platonism in and of itself doesn't have much to say about this, beyond the mere affirmation that there *is* such a realm, that our minds have some sort of access to it, and that this realm is what mathematics is 'about'. So a platonist might respond to our teaser about where there are infinitely many primes with a direct if mysterious answer: 'Within the platonic realm of mathematics.'

This way of thinking about mathematics has some important consequences. One is the idea that mathematics is something we *discover* rather than

something we *invent*. Mathematical concepts and truths are in some way already there, waiting for us to come across them. Furthermore, a mathematical statement (if precisely formulated) is something that is objectively either true or false in the platonic realm, independently of whether we can tell whether it's true. Even if a proof or disproof of, say, the Goldbach conjecture is never discovered by any intelligent being within the lifetime of the universe, there is nonetheless, according to the platonist, a 'fact of the matter' about whether it's true or not.

Historically, platonism has been a very popular view amongst practising mathematicians; it might, perhaps, even be called the 'default' philosophy of mathematics. The great 19th-century French mathematician Charles Hermite expressed this view unequivocally:

> There exists, if I am not mistaken, an entire world which is the totality of mathematical truths, to which we have access only with our mind, just as a world of physical reality exists, the one like the other independent of ourselves, both of divine creation.[22]

And here is a full-on platonist affirmation from a non-mathematical contemporary of Hermite, the American politician and orator Edward Everett:

> In the pure mathematics, we contemplate absolute truths, which existed in the Divine Mind before the morning stars sang together, and which will continue to exist there, when the last of their radiant host shall have fallen from heaven.[23]

A similar and often quoted declaration, but without the religious overtones, comes from the English mathematician (and avowed atheist) G.H. Hardy:

> I will state my own position dogmatically … I believe that mathematical reality lies outside us, that our function is to discover or observe it, and that the theorems which we prove, and which we describe grandiloquently as our 'creations,' are simply the notes of our observations … 317 is a prime number, not because we think it is so, or because our minds are shaped in one way rather than another, but *because it is so*, because mathematical reality is built that way.[24]

Certainly, platonism fits well with what doing mathematics feels like: most practising mathematicians, in some way, have a sense of exploring a pre-existing world. I spoke earlier of mathematics in terms of constructing a building; but the actual experience of doing mathematics often feels more akin to discovering an existing building, just as Emily was doing in our story.

However, if the platonist view has seemed totally reasonable and natural to some, there have equally been others to whom the idea has seemed patently

absurd. To many, the very suggestion of some ‘other world’ in which mathematical truths reside appears ridiculous, fanciful, unbelievable — perhaps even unworthy of serious consideration. As witnesses for such views, I shall cite here a mathematician, a mathematical logician, a pair of cognitive scientists, and a neuroscientist.[25]

> Why do mathematicians believe something so unscientific, so far-fetched as an independent immaterial timeless world of mathematical truth? … Heaven and the Mind of God are no longer heard of in academic discourse. Yet most mathematicians and philosophers of mathematics continue to believe in an independent, immaterial abstract world — a remnant of Plato’s Heaven … Platonists don’t acknowledge the arguments against Platonism. They just reavow Platonism.
>
> — Reuben Hersh, *What is mathematics, really?*

> It will soon be clear to the reader that I am a convinced antiplatonist in mathematics. Briefly, according to the platonist philosophy, the objects of mathematics such as numbers, sets, functions, and spaces are supposed to exist independently of human thoughts and constructions … Though this accords with the mental practice of the working mathematician, I find the viewpoint philosophically preposterous.
>
> — Solomon Feferman, *In the light of logic*

> Finally, there is the issue of whether human mathematics is an instance of, or an approximation to, a transcendent Platonic mathematics. This position presupposes a nonscientific faith in the existence of Platonic mathematics. We will argue that even this position cannot be true … Human beings can have no access to a transcendent Platonic mathematics, if it exists.
>
> — George Lakoff and Rafael Núñez, *Where mathematics comes from*

> It seems to me, to the contrary, that mathematical objects exist in your brain … These “mathematical objects” corresponding to physical states of our brain in such a way that it ought *in principle* to be possible to observe them from the outside looking in … I therefore remain doubtful that mathematical objects exist ‘somewhere in the universe,’ independently of all material and cerebral support, as you would have it.
>
> — Jean-Pierre Changeux, in *Conversations on mind, matter and mathematics*

One also hears platonism variously described as ‘out-of-date’, ‘mystical’, ‘medieval’, ‘untenable’, even ‘unintelligible’. Despite all this, platonism would appear to be alive and well amongst top-flying mathematicians today, if the following quotations are anything to go by:[26]

> I hold … that there exists, independently of the human mind, a raw and immutable mathematical reality … Take prime numbers, for example, which, as far as I am concerned, constitute a more stable reality than the material reality that surrounds us. The working mathematician can be likened to an explorer who sets out to discover the world.
>
> — Alain Connes, in
> *Conversations on mind, matter and mathematics*

> The Mandelbrot set was certainly no invention of any human mind. The set is just objectively there in the mathematics itself. If it has meaning to assign an actual existence to the Mandelbrot set, then that existence is not within our minds … Its existence can only be within the Platonic world of mathematical forms.
>
> — Roger Penrose, *The road to reality*

> Prime numbers would be there regardless of whether we had evolved sufficiently to recognize them … Some philosophers might take issue with such a Platonist view of the world … but to my mind that is what makes them philosophers and not mathematicians.
>
> — Marcus du Sautoy, *The music of the primes*

Obviously, all of this provides the trigger for yet another avalanche of questions. What *are* the arguments for or against platonism? Do the platonists know something the anti-platonists don't, or *vice versa*? Are the two sides somehow talking at cross purposes, or misinterpreting each other's positions? Perhaps there are different kinds of platonism that we need to distinguish, some more credible than others? Or maybe the whole question of whether the platonic realm exists is ultimately meaningless — an illusory dilemma arising from some confusion of language?

If mathematical truths indeed reside in some non-physical platonic realm, is it scientifically credible that the grey porridge in our skulls could have some kind of access to them? Or is the mental image of brains making *contact* with this platonic realm somehow the wrong picture to have?

If we reject platonism, what other views are on offer regarding the 'meaning' of mathematical statements? In what other ways might we understand mathematical statements as saying something both meaningful and true? What is lost, and what is gained, in adopting each of these points of view?

Philosophies of mathematics

What we are really asking, then, through all the specific questions we have touched on, is what is the most viable and satisfactory *philosophy of mathematics* — the best account of what mathematics really is. Ideally, we'd like a philosophy that gave us satisfying answers to all four of the following questions:

1. What kind of meaning (if any) do mathematical statements have?
2. How do we come to have knowledge of mathematical truths (or why does it seem that we do)?
3. In what sense is mathematical knowledge 'certain' and/or 'necessary' (or why does it seem that it is)?
4. Why is mathematics as useful and successful in practice as it is? (Or, if you really insist, why does it seem that way?)

As we have indicated, platonism offers one possible philosophy (or rather class of philosophies: as we'll see later, there are indeed several versions of platonism to be distinguished). But there are also many others to choose from. Some of these emerged from a major period of crisis around the start of the 20th century, when mathematicians discovered that they couldn't agree on the answers to these fundamental questions. The intense and often heated debates of this period gave rise to several major schools of thought, rejoicing in such names as logicism, finitism, intuitionism and predicativism. More recently, other views have joined these on the philosophical market: structuralism, social constructivism, fictionalism. What is more, different philosophies have actually, in some cases, given rise to different brands of mathematics, since certain mathematical ideas — certain principles of reasoning — appear as meaningful and legitimate according to some philosophies but not others.

We have touched on an enormous number of deep questions in this chapter. Needless to say, I won't be able to offer definitive answers to very many of them. The debates between all these schools are still going on, and the basic problems are still unresolved: all of the views mentioned above have their adherents today, and all have their detractors. But I do at least hope to give you a pretty clear idea of what the main issues are. My purpose is to offer a kind of road-map of various possible views and positions, and to introduce you to a range of different ways in which it's possible to think about mathematics, some of which you might never have come across before.

In doing so, I shall try to paint each of these views as vividly and sympathetically as I can, to give you a chance of appreciating how each view feels 'from the inside'. Some of these ways of thinking may seem strange at first, but I hope you'll enjoy having your imagination stretched a bit. Naturally, I come with my own beliefs and prejudices, so I shall probably be

more persuasive with regard to some views than others, simply because they're more congenial to my own mindset. And of course, whatever I write, it will be impossible to satisfy everyone. But I hope, at least, that I will give you enough information to allow you to start reflecting on these issues for yourself.

The mathematics of mathematics

By this point, it might seem as though this is mainly going to be a book about philosophy. Certainly, a lot of the impetus for our explorations will come from the philosophical issues we've been discussing — the big questions concerning the ultimate nature of mathematics. But we are going to approach these questions in a particular way. Namely, we are going to start by asking: What does mathematics itself have to say about mathematics? Can we learn something of philosophical interest from a *mathematical* study of the concepts in question?

This is what the subject of mathematical logic is about. Whereas number theory studies numbers, and geometry studies points, lines and circles, mathematical logic is a branch of mathematics that studies the raw materials of mathematics itself: theorems, proofs, computations, truth. In short, it is *the mathematics of mathematics.*

As we shall see, mathematical logic has a great many interesting and surprising things to reveal about the nature and structure of mathematics. These results don't by themselves provide answers to all the big philosophical questions, but they provide some key pieces of the jigsaw, and they supply the basis for a far more informed and focused philosophical discussion of these questions than would otherwise be possible.

The relevant concepts and results from mathematical logic will take a bit of time to develop, and that is largely what we shall be doing in the first part of the book. In the early chapters of the story, then the focus will be mainly on mathematical and logical ideas, and philosophical issues will play only a minor role — although the commentaries will sometimes expand on the more philosophical aspects. From around Chapter 6 onwards, philosophical questions will start to feature in the story itself, and from there on the mathematics and the philosophy will proceed hand in hand. And as I said in the book's preface, I can promise you that all of this will add up an intellectual adventure of the highest order. The road may be rough at times and the ultimate destination ambiguous at best, but I hope that I've now given you a fair idea of the direction in which we are heading.

In order to make further progress, we shall first need to acquire a little more experience of ordinary mathematics. It is time therefore to return to the Castle to resume our explorations of number theory.

Endnotes

1. In A. Eddington, *The nature of the physical world*, MacMillan (1928).
2. This was undertaken by Tomás Oliveira e Silva and his collaborators. For more information, try an Internet search for 'Goldbach conjecture verification'.
3. For readers who know some calculus, the number of primes less than or equal to *n* is approximated by the *logarithmic integral* function defined by:

$$\text{li}\,(n) \;=\; \int_0^n \mathrm{dx} \,/\, \ln \mathrm{x}$$

4. Littlewood actually proved in 1914 that Gauss's estimate would alternate infinitely many times between 'too high' and 'too low' as *n* got larger and larger. He didn't actually give a numerical estimate for where the number of primes would first overtake the estimate, but in 1955, it was rigorously shown by Stanley Skewes that this must happen before you reach $10^{10^{10^{964}}}$. It's now known that there's a crossover somewhere in the vicinity of 1.397×10^{316}, and it seems likely that this is the first place where the conjecture fails, but to date this has not been rigorously proved. (For further information, see the Wikipedia article on *Skewes' number*.)
5. Interestingly, the idea that mathematics had gone to their heads was picked up by Aristotle: 'The so-called Pythagoreans, who were the first to take up mathematics, not only advanced this subject, but saturated with it, they fancied that the principles of mathematics were the principles of all things.' (Aristotle, *Metaphysics*, Alpha 5.)
6. At the time of writing, there is a wonderful hyperlinked and animated online version of the *Elements*, created by David Joyce. A search for 'Euclid Elements Joyce' will find it.

   The theorem on the infinitude of the primes appears as Book IX, Proposition 20. I've taken a small liberty in the story in ascribing the proof to Euclid himself. It's thought that most of the individual theorems in the *Elements* were known prior to Euclid, and it's very likely that this proof was the work of some earlier mathematician. But it is traditionally known as 'Euclid's proof,' since it's from him that we know it.
7. Euclid himself didn't actually use (the Greek form of) the word *axiom* for his basic assumptions. Rather, his initial premises consisted of five *postulates* (statements specific to the field of geometry) and five *common notions* (statements applicable across a wide range of subjects). In classical Greek usage, the term *axiom* referred strictly speaking to assumptions of the latter kind; in today's usage, we would tend to lump them all together as 'axioms'.
8. This is due largely to the influence of the group of French mathematicians who worked under the collective pseudonym of 'Nicolas Bourbaki', and who practised the axiomatic method in a particularly austere form.
9. Tarski's approach — which is fully up to the most exacting modern standards of precision and rigour — has a rather complicated history. For details of this, and of the axioms themselves, see A. Tarski and S. Givant, *Tarski's system of geometry*. Bulletin of Symbolic Logic, vol. 5, no. 2 (1999).

10. For a fascinating history of the concept of certainty (mathematical or otherwise) from a broad philosophical perspective, see Pasnau (2017).
11. I am grateful to Prof. Jaakko Hämeen-Anttila for drawing this work to my attention, and for helpful correspondence and discussion.
12. Note that what is established as a 'mathematical certainty' is *not* that there was a singularity at the big bang, but merely that *if* general relativity theory is true (and there's enough stuff in the universe) then there must have been such a singularity. As the quotation's context makes clear, Hawking later came to doubt the premise regarding general relativity, but not the mathematical theorem itself.
13. In *The Study of Mathematics* (1902), first published as Chapter 4 of Russell (1910).
14. In *To Infinity and Beyond*, BBC Horizon documentary, February 2010. For much more on multiple universe hypotheses, see Tegmark (2015).
15. Of course, not everyone *would* argue this. The philosopher Descartes argued that one thing you really *couldn't* be wrong about is your own existence. And Wittgenstein argued that in order to doubt anything at all, you implicitly need to assume something else. But we needn't get into these debates here.
16. See for example the account in Kline (1980), which in my opinion seriously overplays the significance of the crises, resulting in a distorted and unduly pessimistic picture of mathematics as a whole.
17. A key text here is R.A. De Millo, R.J. Lipton and A.J. Perlis, *Social processes and proofs of theorems and programs*, Communications of the ACM 22(5) (1979). For a broader discussion of the sociology of proof (including computer-assisted proofs), see MacKenzie (2001).
18. In George and Velleman (2001), p.12
19. In Barrow (1993), p.296. One might be reminded here of Augustine's reflections on another philosophical question: 'What is time? I know very well what it is provided nobody asks me; but if I am asked and try to explain, I am baffled.' (*Confessions*, XI.14)
20. In Plato, *Republic*, IV.510.
21. Some conscientious philosophers nowadays seem either to avoid the term 'platonism', or to reserve it for something more strictly in accord with what Plato himself had in mind. Such philosophers might prefer the term 'realism' for the broader range of views I have in mind here. However, my starting point here is a more informal, 'popular' understanding of platonism which I believe to be widespread among mathematicians.
22. Quoted in Darboux, G., *Éloges académiques et discours*, Paris: A. Hermann et fils (1912); also in Gödel (1995), p.323.
23. Quoted in Bell (1952), p.21. I have not been able to track down the original source for this quotation.
24. In Hardy (1940), §22 and §24.
25. In Hersh (1998), pp.11-13; Feferman (1998), p.ix; Lakoff and Núñez (2000), p.4; Changeux and Connes (1995), pp.13, 14, 18.
26. In Changeux and Connes (1995), pp.26, 12; Penrose (2004), pp.16-17; Du Sautoy (2004), p.7.